

# Joaquín Armijo Torres

*Computational astrophysicist*

## Career Objectives

I am a computational astrophysicist and cosmologist, pioneering simulation-based inference, combining cosmology of the large-scale structure of the Universe with advanced statistical methods to unlock the full scientific potential of current generation surveys such as Rubin-LSST, Roman, and DESI. My research probes both the standard $\Lambda$-CDM model and extensions beyond to explain the accelerated expansion of the Universe. I focus on using information of observed galaxies to link the study of matter, galaxy formation, and the Universe. My methods include the developing of AI-powered techniques to extract maximal information from complex datasets.

## Personal Information

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|---|---|
| Nationality | Chilean |
| Date of Birth | 17th October, 1993 |
| Languages | Spanish (native), English (fluent) |
| ORCID | 0000-0002-5795-7028 |
| SCOPUS | joaquin.armijo@usp.br |

## Skills

**Programming:** Python (NumPy, SciPy, pandas, Matplotlib), C, C++, PyTorch, JAX, CUDA, MPI, OpenMP.
**Advanced Statistics:** Map reconstruction, Simulation-based inference, Bayesian inference, higher-order statistics, emulators.
**Numerical Simulations:** N-body and modified gravity simulations, halo finders, HPC environments, GPU simulations.
**Software Tools:** Git, Linux/Unix, Jupyter, LaTeX, Docker, LLMs.
**Communication:** Teaching and mentoring, outreach talks for large audiences, workshop organization, podcast talks.
**Hobbies:** Sports, storytelling, guitar.

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*https://jarmijotorres.github.io/*

## Software

**CustomCosmologicalAnalysis:** https://github.com/jarmijotorres/CustomCosmologicalAnalysis. Inference tools for multi-messenger cosmological data with theories beyond the standard $\Lambda$CDM.
**GWDALI-Sky**: https://github.com/jarmijotorres/GWDALI-Sky. End-to-end pipeline application of the GWDALI framework for gravitational wave infusion to a custom galaxy observation sky.
**DeepLensingFlow:** https://github.com/jarmijotorres/DeepLensingFlow (Author).
Application of Normalizing flow and Diffusion model for weak lensing statistics (open sourced).
**HOScodes**: https://github.com/LSSTDESC/HOS-Y1-prep/tree/master/hoscodes. Pipeline for LSST-DESC Higher-order statistics topical team. Application of several statistics for the Year-1 Rubin data. (Property of LSST-DESC)

## Education

2018-2022 **Dphil**, *Institute of Computational Cosmology, Durham University*, Durham, United Kingdom, Thesis: *A* framework to test modified gravity using galaxy surveys.

2016 - 2018 **Master in science**, *Institute of Astrophysics, Pontificia Universidad Católica de Chile*, Santiago, Chile., Thesis: *T*esting modified gravity using a marked correlation function..

2012 - 2016 **Degree in Astronomy**, *Pontificia Universidad Católica de Chile*, Santiago, Chile.

## Appointments

### Postdoctoral fellowship.

Rubin-LSST fellow, Departamento de Astronomía. Universidad Técnica Federico Santa María.

October 2026 - present.

### Postdoctoral fellowship.

FAPESP fellow, Institute of Physics, The University of Sao Paulo

October 2025 - September 2026.

### Postdoctoral fellowship.

Project researcher at The Kavli Institute for the Physics and Mathematics of the Universe, The University of Tokyo

October 2022 - September 2025.

### PhD student at the Institute for Computational Cosmology

Supervisors Prof. Carlton Baugh (c.m.baugh@durham.ac.uk), Prof. Peder Norberg

October 2018 - September, 2022.

### Internship at Tharsus Company.

Sponsored by the Centre of Doctoral Training at Durham Univesrity, Tharsus.

Supervisors Dr. Paul Featonby

May - Aug, 2020.

### Visitor astronomer (9 Nights)

European Southern Observatory (ESO) winter internship, Santiago Chile.

Supervisor Prof. Manuela Zoccali and Dr. Luca Sbordone

August, 2014.

## Grants & Awards

### LACEGAL Project internship.

July - August 2017. Pontificia Universidad Católica de Chile. Santiago, Chile

### Programa Formación de Capital Humano Avanzado, becas-Chile 2019. Consejo Nacional de Ciencia Y Tecnología.

October 2018 - September 2022. Durham University, United Kingdom.

### LACEGAL Project internship.

November - December 2019. Institute for Computational Cosmology, Durham University. Durham, United Kingdom.

### Kavli IPMU postdoctoral fellowship.

October 2022 - September 2025. Kavli Institute for the Physics and Mathematics of the Universe, The University of Tokyo. Tokyo, Japan.

### Grants-in-Aid for Scientific Research KAKENHI grant.

March 2023 - February 2025. The Japan Society for the Promotion of Science (JSPS).

### FAPESP fellowship

October 2025 - September 2027. Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP).

### Rubin-LSST fellowship

October 2026 - September 2028. Rubin/Chile fund, AURA-Universidad De Chile.

## Publication list

($\sim 461$ citations; h-index: 8)

**Peer-reviewed publications:**

- **Armijo, J.**, Thiele, L., Liu, J. "Weak lensing statistic covariance replication with flow based models". Phys. Rev. D 114, 023567.
- **Armijo, J.**, Da Costa, L. "Marked statistics across the cosmic web: Environmental dependent clustering in modified gravity simulations". MNRAS, Volume 550, Issue 3, 11 pp.
- Harnois-Déraps, J., Šarčević, N., Medina-Varela, L., **Armijo, J.** et al. "Non-linear infusion of intrinsic alignment and source clustering: impact on non-Gaussian cosmic shear statistics". MNRAS, Volume 550, Issue 2, 24 pp.
- Tokiwa, A. Bayer, A. E., **Armijo, J**, Liu, J., et al. "Impact of Simulation Box Size for Weak Lensing: Replication and Super-Sample Effects". JCAP, Volume 2026, Issue 07, 38 pp.
- Cowell, J. A. **Armijo, J.**, Thiele, L., Marques, G. A., Novaes, C. P., Grandón, D., Cheng, S. Shirasaki, M., Alonso, D. & Liu, J., "First Constraints from Marked Angular Power Spectra with Subaru Hyper Suprime-Cam Survey First-Year Data " MNRAS, Volume 546, Issue 2, 17 pp.
- **Armijo, J.**, Marques, G. A., Novaes, C. P., Thiele, L., Cowell, J. A., Grandón, D., Shirasaki, M., & Liu, J., "Cosmological constraints using Minkowski functionals from the first year data of the Hyper

Suprime-Cam",Monthly Notices of the Royal Astronomical Society, Volume 537, Issue 4, March 2025, Pages 3553–3560, (2025).
- Novaes, C. P., Thiele, L., **Armijo, J.**, et al., "Cosmology from HSC Y1 Weak Lensing with Combined Higher-Order Statistics and Simulation-based Inference", Phys. Rev. D 111, 083510 (2025).
- **Armijo, J.**, Baugh, C. M., Norberg, P., & Padilla, N. D., "A new test of gravity - I. Introduction to the method", Monthly Notices of the Royal Astronomical Society, 529, 2866-2876, (2024).
- **Armijo, J.**, Baugh, C. M., Norberg, P., & Padilla, N. D., "A new test of gravity - II. Application of marked correlation functions to luminous red galaxy samples", Monthly Notices of the Royal Astronomical Society, 528, 6631-6636, (2024).
- **Armijo, J.**, Baugh, C. M., Padilla, N. D., Norberg, P., & Arnold, C., "Making use of sub-resolution haloes in N-body simulations", Monthly Notices of the Royal Astronomical Society, 510, 29-33, (2022).
- Paillas, E., Cautun, M., Li, B., Cai, Y.-C., Padilla, N., **Armijo, J.**, & Bose, S., "The Santiago-Harvard-Edinburgh-Durham void comparison II: unveiling the Vainshtein screening using weak lensing", Monthly Notices of the Royal Astronomical Society, 484, 1149-1165, (2019).
- **Armijo, J.**, Cai, Y.-C., Padilla, N., Li, B., & Peacock, J. A., "Testing modified gravity using a marked correlation function", Monthly Notices of the Royal Astronomical Society, 478, 3627-3632, (2018).
- Cautun, M., Paillas, E., Cai, Y.-C., Bose, S., **Armijo, J.**, Li, B., & Padilla, N., "The Santiago-Harvard-Edinburgh-Durham void comparison - I. SHEDding light on chameleon gravity tests", Monthly Notices of the Royal Astronomical Society, 476, 3195-3217, (2018).

## Workshops and Seminars

### Contributed talks:

2016 **Second Chilean Workshop on Theoretical and Numerical Astrophysics.** 4 - 5th August, Pontificia Universidad Católica de Chile. Santiago, Chile.

2017 **7th Friends of Frieds Meeting.** 3 - 7th April. Observatorio Astronómico Nacional. Córdoba, Argentina.

2017 **Galaxy Groups and Clusters II: Laboratories to study galaxy evolution.** October 23-26. La Serena, Chile.

2017 **Southern Astrophysics Network. November 9-10. Universidad Andrés Bello.** Santiago, Chile.

2018 **CosmoAndes. January. Pontificia Universidad Católica.** Santiago, Chile.

2018 **SDSS group meeting. February. MPE Garching, Germany.**

2018 **National CDT meeting. November. The University of Edinburgh.** Edinburgh, United Kingdom.

2019 **The Physics of the Accelerating Universe Survey collaboration meeting. March.** Barcelona, Spain.

2019 **Mock Córdoba. April. Instituto de Astronomía Teórica y Experimental.** Santiago, Chile.

2019 **CDT summer school. The University of Sussex. July.** Brighton, United Kingdom.

2019 **DESI-UK collaboration meeting. University College London. October.** London, United Kingdom.

2020 **Mock Innsbruck. March. Innsbruck University.** Innsbruck. Austria.

2021 **DEX Virtual Meeting. January. Durham - Edinburgh (virtual).**

2023 **CMB $\times$ LSS at Kyoto University. April.** Kyoto. Japan.

2023 **Astro AI with Fugaku at The University of Tokyo. September.** Tokyo. Japan.

2023 **2nd Shanghai Assembly on Cosmology and Structure Formation. November. SJTU.** Shanghai. China.

2024 **CD3$\times$CCA joint workshop. January. Kavli IPMU.** Kashiwa. Japan.

2024 **COSMO21 Astrostatistics. May.** Chania. Greece.

2024 **LSS Quest at Osaka University. July.** Osaka. Japan.

2024 **COSMO2024**. Kyoto. Japan.

2025 **Scientific AI for Physics Workshop**. São Paulo. Brazil

2026 **GR Effects in LSS**. Aegina. Greece.

2026 **Cosmic Rhythms**. Rio de Janeiro. Brasil.

2026 **CosmoSur VII**. Valparaiso. Chile.

Invited talks:

2022 **IfA Seminar at University of Edinburgh. September.** Edinburgh. United Kingdom.

2023 **Seminar at UTFSM. March.** Santiago. Chile.

2024 **Seminar at PUCV. April.** Valparaiso. Chile.

2024 **Observatorio Calán at U. de Chile seminar. April.** Santiago. Chile.

2024 **Mock Barcelona at UAB. October.** Barcelona. Spain.

2025 **LeCosPA meets IPMU. March.** Taipei. Taiwan.

2025 **Friday Seminar at ICC, Durham University.** Durham. UK.

2025 **Workshop on Large-scale Structure Oxford-IPMU at Oxford University.** Oxford. UK.

2025 **ICTP seminar UNESP-USP.** São Paulo. Brazil.

2025 **Agora Forum.** São Paulo. Brazil.

2026 **Astronomy Seminar at USM.** Santiago. Chile.

2026 **Astronomy Seminar at UDP.** Santiago. Chile.

## Extra-Curricular Activities and Popular Science

2013-2018 Planetarium Talks. Instituto de Astrofísica, Instituto Milenio de Astrofísica.

2015-2017 Manuel Foster Observatory. Monitor guide.

2018 Astro-informatics interviewer qualitative focus project

2019-2022 Science outreach: Celebrate science, Durham University.

2024 Citizen science outreach talks: My life as an astronomer in Japan. Kashiwa city hall.

2025 Citizen science outreach talks: From small to large: The size and scale of astronomical objects. Tamarokuto Science Center. Tokyo.

2025 Podcast on science: The den of the Unknown: The cosmic ladder.

2026 Reels AstroGainz: Research presentation.

## Leadership

2022 CDT Durham virtual summer school. SOC/LOC.

2023 Astro Journal club organizing. Kavli IPMU.

2024 CD3 group meetings and hack fridays. Kavli IPMU.

2024 CD3×CCA joint workshop. Kavli IPMU. SOC hackton organizer.

2024 Baryons in the Universe April. Kavli IPMU. LOC.

2025 Beyond two-point statistics meets survey systematic workshop, September. Kavli IPMU. LOC.

2025 Beyond two-point statistics meets survey systematic workshop, September. Kavli IPMU. LOC.

2026 Satellite cosmology workshop, September. Instituto Principia. co-Organizer.

## Mentoring

2025 Lucas Da Costa. ILANCE. Project: The environmental dependent clustering in modified gravity simulations.

2024 Valentin Clarisse. ILANCE. Project: The information inside marked correlation function and power spectrum in the Quijote-MG simulations.

2024 Akira Tokiwa. Kavli IPMU. Project: Super sample covariance in weak lensing higher-order statistics.

2024 Jess Cowell. Kavli IPMU. Project: First application of the marked power spectrum on the HSC-Y1 data.

2026 Roberta Vasallo. IFUSP. Project: Revealing Assembly bias using the marked correlation function in DESI-DR1.

## Teaching Training

### Lectures

2026 Topics in astrophysics. Weak lensing and simulation based inference. UNAB graduate course. Speaker. Organized by Prof. Celeste Artale

2025 Cosmology and large-scale structure crash course for ILANCE students. Organizer, Organized by Prof. Elisa Ferreira.

### Teacher's assistant

2025 Handling Large-scale structure numerical simulations in python. Crash course for master and PhD students.

2021-2022 Python programming introduction for PhD students. Durham University

### Teacher's assistant

2016-2017 Space-time and Universe. Professor: Dr. Rolando Dünner. Pontificia Universidad católica de Chile

2015 Classical Physics. Professor: Dr. José Mejía. Pontificia Universidad Católica de Chile

2012-2017 Classical Mechanics. Facultad de Física. Pontificia Universidad Católica de Chile.